CS103: Project Topics

If no links are provided for the topics below, please conduct your own web research to get started. **Wikipedia** is often a good place to begin your exploration.

1. **Linear Programming and Optimization (Solver in Excel)**
    - **Hint**: Start by understanding the geometry of linear programming problems. Solver in Excel is a great tool for visualizing feasible regions and optimal solutions. Make sure to analyze how constraints impact the solution.
    - **Further Exploration**: Try solving real-world optimization problems, such as resource allocation or scheduling.
    - **Simple Explanation**: https://youtu.be/K7TL5NMlKIk
    - **Wikipedia**: https://en.wikipedia.org/wiki/Simplex_algorithm
2. **Neural Network Expressed in Linear Algebra**
    a. **Hint**: Neural networks can be expressed as matrix operations where weights and biases are adjusted through backpropagation. Focus on how the dot product and matrix multiplication are used in forward propagation.
    b. **Further Exploration**: Implement a simple neural network and experiment with different activation functions.
    c. Neural Net in Matrix Form
    d. https://youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi (3B1B)
3. **Image Processing: Filters**
    a. **Hint**: Filters like edge detection (Sobel, Prewitt) and Gaussian blur can be implemented by convolving an image matrix with a filter matrix. Understand how different filters modify image features.
    b. **Further Exploration**: Test how different kernel sizes impact the result and implement your own custom filters.
    c. https://www.nibcode.com/en/blog/1136/linear-algebra-and-digital-image-processing-part-II-filters
4. **Keystone/Perspective Correction**
    a. **Hint**: Perspective correction uses projective transformations, which can be represented by a 3x3 matrix. Experiment with transforming images and correcting distortions caused by perspective.
    b. **Further Exploration**: Try applying this to a real-world photo and use matrix multiplication to fix distorted elements.
    c. https://mathematica.stackexchange.com/questions/121822/application-of-the-projection-matrix-removing-perspective-distortion
5. **Kirchhoff's Laws**
    a. **Hint**: Kirchhoff's current and voltage laws can be solved using systems of linear equations. Set up the equations from a given circuit and use matrix methods to solve for unknowns.
    b. **Further Exploration**: Experiment with solving more complex circuits that include resistors, capacitors, and inductors.
    c. **Videos**:
        1. Gilbert Strang: https://youtu.be/Q95lUJagN0A
        2. Catherine Schmurr: https://youtu.be/LVDX7T-jX3I
6. **Fourier Transform**
    a. **Hint**: Fourier Transforms break down signals into their frequency components. Pay special attention to how matrix multiplication is used to perform these transformations.

b. **Further Exploration**: Implement the Discrete Fourier Transform (DFT) for simple signals and compare the results with Fast Fourier Transform (FFT).
c. Videos
    1. Gilbert Strang: https://youtu.be/lGGDIGizcQ0?t=2112
    2. 3B1B: https://youtu.be/spUNpyF58BY

7. Fast Fourier Transform (FFT)

a. **Hint**: The FFT is an optimized algorithm for computing the Fourier Transform, significantly reducing computational complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$. Focus on understanding how the divide-and-conquer approach is employed to achieve this efficiency
b. **Further Exploration**: Compare the performance of DFT and FFT in terms of speed for larger datasets.
c. **Gilbert Strang**: https://youtu.be/M0Sa8fLOajA

8. Data Compression Using SVD

a. **Hint**: Singular Value Decomposition (SVD) is used for data compression by reducing matrix rank while preserving essential features. For this project, compress an image and analyze the trade-off between compression and quality.
b. **Further Exploration**: Experiment with different numbers of singular values and compare image quality across different datasets.
c. https://dustinstansbury.github.io/theclevermachine/svd-data-compression
d. **Toby Driscoll**: https://youtu.be/QP4hv5cRUiI

9. Principal Component Analysis (PCA) from SVD

a. **Hint**: PCA reduces the dimensionality of data while retaining variance. Implement PCA using SVD to extract principal components from a dataset and visualize how much variance is explained by each component.
b. **Further Exploration**: Apply PCA to a high-dimensional dataset, such as a facial recognition dataset, to explore its effectiveness in feature reduction.
c. Alexander Ihler: https://youtu.be/F-nfsSq42ow

10. K-Means Clustering

a. **Hint**: K-Means clustering partitions data into clusters based on the distance between data points. Try implementing K-Means using matrix operations and analyze how the clusters change with different initial centroids.
b. **Further Exploration**: Experiment with different distance metrics, such as Manhattan distance, to see how they affect clustering.
c. Stanford Engineering, Stephen Boyd: https://youtu.be/liaZ_SCuE1w
d. https://arxiv.org/pdf/1512.07548.pdf

11. Covariance matrix and VRC

a. **Hint**: The covariance matrix describes the relationship between different variables in multivariate data. The Variance Ratio Criterion (VRC) is a clustering validity index that uses covariance information. Explore how the covariance matrix impacts clustering.
b. **Further Exploration**: Calculate and interpret the covariance matrix for different datasets to understand variable relationships.
c. Start here: http://www.lab.upc.edu/software/codas/vrc_clustering.html
d. Consult the prof and get another paper

12. Simulation of Multivariate Normal Distribution with a Given Covariance Matrix

a. **Hint**: Simulating multivariate normal distributions involves generating random vectors from a covariance matrix. Use matrix decomposition techniques such as Cholesky decomposition for this simulation.
b. **Further Exploration**: Experiment with different covariance structures and observe how they affect the shape and spread of the data.

c. https://juanitorduz.github.io/multivariate_normal/

13. **Vector Space Model and Cosine similarity in Text Analytics**
    a. **Hint**: Cosine similarity measures the angle between two vectors in high-dimensional space. It is particularly useful in text analytics for comparing document similarity. Implement cosine similarity using dot products and magnitude calculations.
    b. **Further Exploration**: Try using cosine similarity to compare multiple documents and observe how similarity scores change with different text corpora.
    c. https://www.machinelearningplus.com/nlp/cosine-similarity/
14. **Logistic Regression and Polynomial Regression**
    a. **Hint**: Logistic regression is used for binary classification, and polynomial regression models non-linear relationships. Implement logistic regression using matrix methods and experiment with polynomial terms for regression models.
    b. **Further Exploration**: Test how adding more features or higher-order terms affects model accuracy and overfitting.
    c. https://www.jeremyjordan.me/logistic-regression/
    d. https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/what-is-logistic-regression/
    e. https://www.analyticsvidhya.com/blog/2021/07/all-you-need-to-know-about-polynomial-regression/

Additional Topics and Helpful Hints:

The following topics are generated by ChatGPT.

15. **Markov Chains and Transition Matrices**
    a. **Hint**: Markov chains use transition matrices to describe systems that change over time. Simulate a Markov process and analyze its long-term behavior using matrix powers.
    b. **Further Exploration**: Use Markov chains to model real-world phenomena such as weather changes or stock market states.
16. **Graph Theory and Adjacency Matrices**
    a. **Hint**: Adjacency matrices represent connections between nodes in a graph. Implement algorithms such as Breadth-First Search (BFS) or Depth-First Search (DFS) using matrix operations.
    b. **Further Exploration**: Investigate how eigenvalues and eigenvectors of the adjacency matrix can be used to detect clusters in the graph.
17. **Linear Algebra in Machine Learning**
    a. **Hint**: Many machine learning algorithms rely on linear algebra for optimization, such as Gradient Descent. Implement Gradient Descent and explore how matrix operations are used to update model parameters.
    b. **Further Exploration**: Apply this to a linear regression model and investigate convergence rates based on different learning rates.
18. **Linear Transformation and Computer Graphics**
    a. **Hint**: Linear transformations can represent rotations, translations, and scalings in computer graphics. Visualize how these transformations affect 2D or 3D shapes.
    b. **Further Exploration**: Implement 3D perspective transformations and analyze how they affect the appearance of objects.

Not Recommended

If you select any of these topics that are not recommended, please consult the prof first.

19. Google Page Rank algorithm
    a. Week 12 Lab
    b. M4ML: https://youtu.be/urKLHNhUEQ0
    c. Consult prof first: Not recommended because it is discussed in class videos
20. Sparse Matrices
    a. Wikipedia: https://en.wikipedia.org/wiki/Sparse_matrix
    b. Lab from the prof
    c. Consult prof first: Not recommended because already done in the lab
21. Linear algebra in Special Relativity
    a. Sean Johnson's Notes
    b. Not recommended: Too much physics
22. Linear Algebra for Quantum Mechanics
    a. BalKi (the prof's prof): https://youtu.be/y3ARLfm-52w?t=471
    b. Not recommended: Too much physics

Your Own Topics

In addition to the topics suggested above, you can come up with your own topics for your project, but please consult the prof first.

Notes

1. The "**Hint**" and "**Further Exploration**" were generated by ChatGPT and are listed here for your consideration. You are not required to follow them unless you find them helpful.
2. The links provided serve as starting points for your exploration.
3. When presenting, remember that your classmates (and possibly the teaching team) may not be familiar with your chosen topic:
    a. Define key terms.
    b. Focus on clarity and effective information transfer.
    c. Avoid overwhelming the audience with complex mathematics or programming.
    d. Ensure you fully understand the material you are presenting
4. All the best!